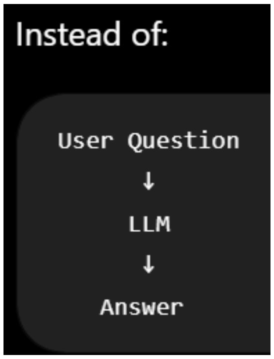
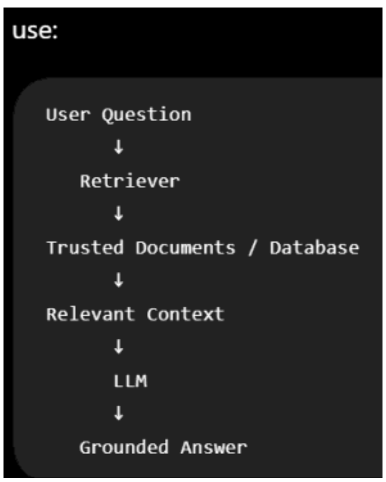
Hallucination Control

Wednesday, July 29, 2026 7:55 PM

1. Use RAG - Ground the LLM with real data





Ex: What is the warranty period for Product X?

The LLM should retrieve the actual warranty document rather than relying on its pre-trained knowledge.

2. Improve retrieval quality

Bad retrieval -> bad context -> hallucinated answer.

Use:

- Good chunking strategy
- Appropriate embedding model
- Hybrid search: Keyword + vector
- Reranking
- Top-K tuning
- Remove duplicate / Irrelevant chunks

For example:

```
Question
   ↓
Vector Search
   ↓
Top 20 chunks
   ↓
Reranker
   ↓
Best 5 chunks
   ↓
LLM
```

3. **Strong prompting:**

Tell the model explicitly:

```
Answer only using the provided context.

If the answer cannot be found in the context,
say "I don't have enough information to answer."

Do not make assumptions.
Do not invent facts.
```

This helps, although **prompting alone cannot eliminate hallucinations.**

4. **Use temperature appropriately**

For factual applications:

```
temperature = 0
```

or a low value such as:

```
0.0 – 0.3
```

Generally, lower temperature reduces randomness.

But remember:

> **Temperature = 0 does NOT guarantee zero hallucination.**

5. **Give the LLM authoritative sources**

Prefer:

```
Database
Official documentation
Verified knowledge base
Approved enterprise documents
```

over:

```
Random web pages
Unverified documents
User-generated content
```

The quality of the source directly affects the quality of the answer.

6. **Add citations / source attribution**

For RAG applications, return:

```
Answer:
The warranty period is 2 years.

Sources:
- Warranty_Policy.pdf
- Section 4.2
```

This makes the response **traceable and auditable.**

7. Use structured output

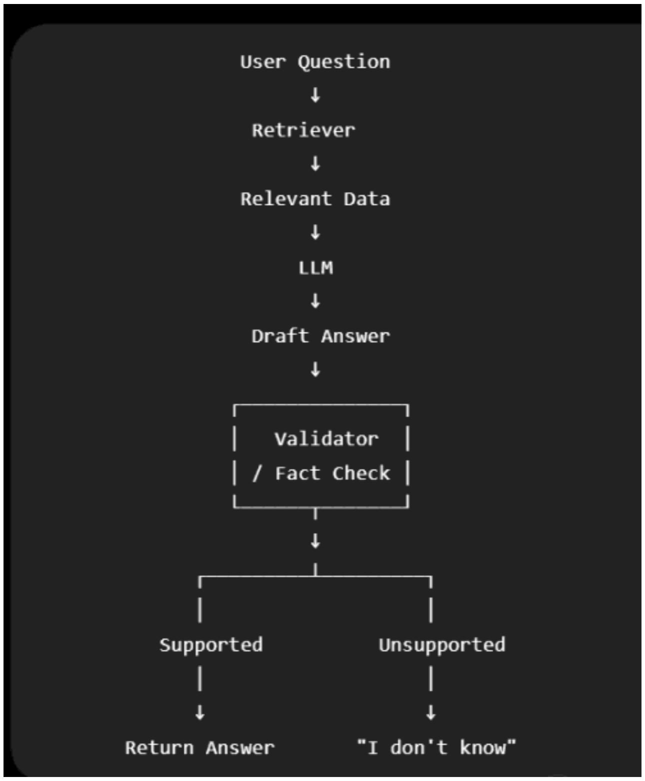
Instead of allowing completely free-form output:

JSON

```json
{
  "answer": "...",
  "confidence": 0.92,
  "sources": [
    "document_123"
  ]
}
```

You can validate whether required fields are present.

8. Add a verification layer



Summary:
How do you prevent hallucinations?
RAG
Prompt Engineering
Context Restriction
Source citations
Temperature tuning
Validation Layer
Confidence scoring
Human Approval